

Retrieval-Augmented Generation (RAG) – MNLP HW2

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1 Introduction

We address short-answer factual generation with small open-weight language models. Starting from the dense retriever developed in Homework 1, we compare generation without context, standard Retrieval-Augmented Generation (RAG), and oracle RAG. The goal is to understand how much retrieved evidence helps small LMs, how reliable the automatic evaluation is, and whether additional prompting or knowledge enrichment improves the final answers. We evaluate Qwen3-0.6B and SmolLM2-360M-Instruct, and extend the base system with compressed/corrective prompting (A1), alternative correctness judges (A2), and Wikidata-based context enrichment (A3).

2 Methodology

For each question, we rank candidate chunks with the fine-tuned all-MiniLM-L6-v2 retriever and select the top-3 passages. In the baseline setting, the generator receives only the question. In standard RAG, the retrieved chunks are prepended to the question. In oracle RAG, the gold chunk is forced into the first context position, while keeping the same generation pipeline; this isolates the effect of retrieval quality and context ordering from the intrinsic generation ability of the model. All prompts ask for short factual answers, and generated outputs are post-processed by removing answer prefixes and keeping the first non-empty line.

Evaluation combines three levels. First, we compute automatic metrics: Exact Match (EM), substring Exact Match (subEM), and METEOR. Second, we use SmolLM2 itself as the original LLM-as-a-Judge on a 200-sample subset, assigning a binary correctness label according to whether the generated answer contains the gold short answer. Third, both group members manually annotate the same subset and we compute Cohen’s Kappa between humans and automatic judges. This is im-

portant because the original judge can be biased toward accepting related but incorrect answers.

For A1, Compressed RAG reduces context noise before generation: each retrieved chunk is split into sentences, the query and candidate sentences are embedded with MiniLM, cosine similarity is used to select the most relevant sentences, and their original order is preserved in the compressed prompt. Corrective RAG instead modifies the prompt to make the model answer only when the retrieved evidence is sufficient. For A2, we compare the original SmolLM2 judge with Qwen2.5-3B-Instruct and Qwen2.5-1.5B-Instruct. For A3, we enrich the prompt with Wikidata labels and descriptions derived from the provided entity identifiers.

3 Experimental Setup

Experiments are run on the MNLP 2026 Homework 2 dataset. We evaluate Qwen3-0.6B and SmolLM2-360M-Instruct on the 2,000-example test split and produce the required blind-set JSONL outputs. For each model, we compare three settings: baseline, standard RAG, and oracle RAG. Standard RAG uses the fine-tuned MiniLM retriever with top-3 chunks, while oracle RAG forces the gold evidence to appear first in the context.

We evaluate generations with EM, subEM, and METEOR. In addition, we manually validate 200 outputs from the best standard RAG model, SmolLM2-360M-Instruct, comparing human annotations with the original SmolLM2 judge and the A2 alternative judges. We also perform a qualitative failure analysis to separate judge errors, generation failures, retrieval failures, and human disagreements.

For A1, we test two variants with SmolLM2-360M: Compressed RAG, which uses MiniLM sentence embeddings and cosine similarity to keep only the most query-relevant sentences, and Corrective RAG, which uses a stricter prompt requiring

sufficient contextual evidence. For A3, we enrich the prompt with Wikidata label and description associated with the provided entity identifier, testing whether compact external entity knowledge improves factual answering.

4 Results and Discussion

Table 1 reports the main generation results. Both models benefit strongly from retrieved context: baseline scores are very low, while RAG improves all metrics. Oracle RAG is consistently stronger than standard RAG, showing that retrieval quality and evidence ordering still limit the system. SmolLM2-360M-Instruct is the best standard RAG model, reaching 15.85 EM, 32.25 subEM, and 32.88 METEOR. Despite being smaller than Qwen3-0.6B, it performs better in our setting, likely because its instruction tuning and answer format are better aligned with the task.

Model	Setup	EM	subEM	METEOR
Qwen3	Baseline	0.30	2.15	5.41
Qwen3	RAG	5.65	29.75	30.63
Qwen3	Oracle	6.70	33.75	34.33
SmolLM2	Baseline	0.75	6.15	8.96
SmolLM2	RAG	15.85	32.25	32.88
SmolLM2	Oracle	16.90	34.05	34.62

Table 1: Main generation results on the test split.

Table 2 summarizes the manual validation. The two human annotators almost perfectly agree ($\kappa = 0.9898$), while the original SmolLM2 judge has near-zero agreement with humans and marks 96.5% of the answers as correct.

Evaluator	Positive Rate	κ vs H1	κ vs H2
Human annotators	56.50%	0.9898	
SmolLM2 Judge	96.50%	0.0672	0.0688

Table 2: Manual validation and original LLM-as-a-Judge agreement on 200 samples. The SmolLM2 judge shows a very high positive rate and very low agreement with both human annotators.

The failure analysis in Table 3 confirms this issue: 80 out of 200 samples are judge false positives. Therefore, the original LLM-as-a-Judge is too permissive and cannot be used alone as a reliable correctness signal. In contrast, Qwen2.5-1.5B has a more realistic positive rate and much higher agreement with both humans ($\kappa \approx 0.81$), showing that judge choice strongly affects evaluation reliability.

Error type	Count	% failures	% samples
Judge false positive	80	90.91	40.0
Generation failure	4	4.55	2.0
Retrieval failure	2	2.27	1.0
Human disagreement	1	1.14	0.5
Judge false negative	1	1.14	0.5

Table 3: Failure analysis on the 200 manually annotated samples.

Judge	Pos. Rate	κ vs H1	κ vs H2
Original judge	96.5	0.0672	0.0688
Qwen2.5-3B	38.5	0.5728	0.5640
Qwen2.5-1.5B	53.5	0.8183	0.8081

Table 4: A2 agreement comparison between the original judge, alternative Qwen judges, and human annotators.

The additional experiments refine the interpretation. Compressed RAG is the strongest A1 variant, improving EM from 15.85 to 17.80 and METEOR from 32.88 to 33.33, suggesting that sentence-level compression helps remove distracting context. Corrective RAG is more conservative, but does not improve exact-answer quality.

Configuration	EM	subEM	METEOR
Standard RAG	15.85	32.25	32.88
Compressed RAG (A1)	17.80	32.60	33.33
Corrective RAG (A1)	15.60	32.80	32.36

Table 5: A1 prompting strategy comparison for SmolLM2-360M-Instruct on the test split.

Wikidata enrichment has only a marginal effect: subEM and METEOR slightly improve, but EM decreases from 15.85 to 15.75, suggesting that external entity descriptions can help lexical overlap while also adding prompt noise. Overall, RAG is essential for small models, oracle results show that retrieval still matters, and manual validation shows that automatic judges must be carefully calibrated.

Configuration	EM	subEM	METEOR
Standard RAG	15.85	32.25	32.88
Wikidata-Enriched RAG	15.75	32.70	33.06

Table 6: A3 Wikidata-enriched RAG results for SmolLM2-360M-Instruct on the test split.